EEG-Based Epilepsy Detection Using Multi-Band Feature Extraction Techniques and Machine Learning

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Abstract—Epilepsy is a chronic neurological disorder characterized by recurrent, unprovoked seizures resulting from abnormal, excessive, and synchronized electrical discharges in brain neurons. Because visual inspection of long-term electroencephalography (EEG) recordings is time-consuming and subject to inter-observer variability, automated and interpretable diagnostic pipelines are critical for clinical decision support. This study presents an end-to-end multi-band feature extraction and machine learning framework for automated epileptic seizure detection using primary clinical EEG data. Over a three-month clinical data acquisition period at the Medicine Department of Khulna Medical College and Hospital (Bangladesh), raw 19-channel scalp EEG recordings were collected from 51 patients and curated into a balanced cohort of 32 subjects (16 seizure-positive patients and 16 seizure-negative controls). Raw Neuro-Scan (.eeg) signals were converted to European Data Format (.edf) and preprocessed in MATLAB (EEGLAB 2021.1) using bandpass filtering across three neurophysiologically distinct frequency bands: Alpha (8–13 Hz), Beta (14–29 Hz), and Gamma (30–50 Hz). Six statistical time-domain biomarkers—Entropy, Kurtosis, Coefficient of Variation (CV), Hjorth Parameters (Activity, Mobility, Complexity), Signal Energy, and Skewness—were extracted alongside 2D time-frequency spectrograms. Using an 80:20 train-test split and systematic GridSearch hyperparameter optimization, Random Forest (RF) and Decision Tree (CART) classifiers were evaluated on each individual feature across all three frequency bands, while a 2D Convolutional Neural Network (CNN) was trained on spectrogram representations. Experimental validation demonstrates that hyperparameter-tuned Random Forest models achieve peak classification accuracies of 90.0% on Alpha-band features (ROC-AUC = 0.95) and 90.0% on Gamma-band Energy Axial features (ROC-AUC = 0.83), as well as 89.0% accuracy on Beta-band Skewness features (ROC-AUC = 0.93), outperforming standalone Decision Tree models (84%–88%) and 2D CNN spectrogram baselines (75.0%).

Index Terms—Electroencephalography (EEG), Epileptic Seizure Detection, Multi-Band Bandpass Filtering, Hjorth Parameters, Time-Frequency Spectrograms, Random Forest, Decision Tree, Convolutional Neural Network (CNN).

I. INTRODUCTION

Approximately 50 million individuals worldwide live with epilepsy, a chronic non-communicable neurological disorder characterized by recurrent, unprovoked seizures resulting from excessive and hypersynchronized electrical discharges within cerebral neuronal networks [1], [8]. Beyond the immediate physical risks of tonic-clonic convulsions, falls, and transient loss of consciousness, the unpredictability of seizure onset imposes severe psychological, cognitive, and socioeconomic burdens on patients and caregivers [17], [32]. While anticonvulsant pharmacotherapy and surgical interventions exist, early and objective clinical diagnosis remains essential for personalized treatment planning.

Electroencephalography (EEG) serves as the clinical gold standard for capturing real-time neurophysiological dynamics with millisecond temporal resolution via scalp electrodes [3], [25]. Unlike structural neuroimaging modalities such as Magnetic Resonance Imaging (MRI) or Positron Emission Tomography (PET), multi-channel EEG directly records post-synaptic cortical potentials across distinct brain regions. However, conventional clinical EEG evaluation requires extended monitoring sessions (often 1.5 hours or longer) and relies heavily on visual inspection by specialized neurologists. This manual workflow is labor-intensive, subjective, and susceptible to ocular, muscular, and environmental artifacts [13].

While recent machine learning (ML) and deep learning studies have demonstrated high classification accuracy on benchmark repositories, two practical gaps remain in clinical EEG research: (1) the scarcity of primary, hospital-acquired EEG datasets from underrepresented South Asian clinical populations, and (2) the tendency of existing studies to concatenate all extracted features into a single opaque vector without isolating the discriminative contribution of individual statistical biomarkers across specific neural frequency bands. To address these challenges, this paper makes three primary contributions:

1. **Primary Clinical EEG Dataset Acquisition:** We conducted a three-month clinical data collection initiative at the Medicine Department of Khulna Medical College and Hospital (Khulna, Bangladesh), screening 51 clinical referrals to curate a balanced 32-subject, 19-channel raw EEG cohort (16 epileptic seizure-positive patients and 16 non-epileptic controls).

2. **Multi-Band Spectral Decomposition Pipeline:** We designed a standardized preprocessing workflow converting proprietary Neuro-Scan (.eeg) clinical files into European Data Format (.edf) and applying zero-phase bandpass filtering in MATLAB (EEGLAB 2021.1) across Alpha (8–13 Hz), Beta (14–29 Hz), and Gamma (30–50 Hz) rhythms.

3. **Granular Feature-by-Feature & Multi-View Benchmarking:** Rather than aggregating all biomarkers blindly, we systematically evaluate six distinct time-domain features (Entropy, Kurtosis, Coefficient of Variation, Hjorth Parameters, Energy, and Skewness) individually alongside 2D time-frequency spectrograms across hyperparameter-tuned Random Forest, Decision Tree, and 2D CNN architectures.

II. RELATED WORK

Automated epileptic seizure detection from multi-channel EEG signals has evolved across spectral, statistical, and deep learning paradigms. Early approaches focused on dimensionality reduction and neural synchrony. Principal Component Analysis (PCA) and Robust-PCA (R-PCA) have been widely paired with Support Vector Machines (SVM), Naive Bayes, and k-Nearest Neighbors (k-NN) to isolate seizure-related covariance structures from noisy scalp recordings [5], [7], [20]. For instance, R-PCA feature extraction combined with Gaussian SVM achieved 93.18% accuracy on a 13-subject custom EEG cohort, substantially outperforming raw-matrix k-NN baselines (52.27%) [20].

Time-domain and wavelet-domain biomarkers have also proven highly effective for tree-based and ensemble classifiers. Chiang et al. [4] integrated discrete wavelet transforms (DWT) with fuzzy entropy and associative Petri nets, while Sharmila and Geethanjali [12] utilized DWT statistical features with k-NN and Naive Bayes classifiers. Mehrnoosh et al. [10] decomposed EEG signals via DWT and empirical mode decomposition (EMD), extracting variance, skewness, kurtosis, Shannon entropy, and Hjorth parameters to classify healthy controls and epileptic cases using SVM and Regularized Linear Discriminant Analysis (RLDA). Similarly, tree-based supervised frameworks using surrogate Decision Trees and Random Forests have demonstrated high interpretability and strong generalization across multi-channel montages [8], [13].

More recently, multi-view Convolutional Neural Networks (CNNs) and hybrid CNN-BiLSTM architectures have been introduced to learn joint time-frequency representations from EEG spectrograms [9], [14], [15], [27], [28]. Liu et al. [9] proposed a multi-view CNN combining time-domain and frequency-domain views for seizure prediction, while Malekzadeh et al. [27] fused handcrafted statistical features with deep representations. However, end-to-end 2D CNN models typically require massive labeled training sets to avoid overfitting, whereas ensemble tree classifiers (such as Random Forests) trained on band-specific statistical biomarkers often provide superior stability and diagnostic transparency on small-to-medium clinical hospital cohorts.

III. CLINICAL DATASET ACQUISITION & MULTI-BAND PREPROCESSING

A. Clinical Cohort & 19-Channel EEG Acquisition Protocol

Primary clinical EEG data were collected over a three-month period at the Medicine Department of Khulna Medical College and Hospital, Khulna, Bangladesh. All recordings were acquired under clinical supervision using a 19-channel clinical Neuro-Scan EEG system arranged according to the international 10–20 electrode placement standard: Fp1-A1, Fp2-A2, F3-A1, F4-A2, C3-A1, C4-A2, P3-A1, P4-A2, O1-A1, O2-A2, F7-A2, F8-A2, T7-A1, T8-A2, P7-A1, P8-A2, Fz-A1, Pz-A2, and Cz-A2. Each clinical diagnostic session lasted approximately 1.5 hours, capturing resting-state, eyes-open, and eyes-closed neurological dynamics alongside clinician-verified diagnostic reports.

From an initial pool of 51 examined patients, incomplete or heavily corrupted recordings were excluded to establish a strictly balanced, noise-verified cohort of 32 subjects: 16 confirmed epileptic seizure-positive patients (Class 1) and 16 non-epileptic control subjects (Class 0). Proprietary clinical Neuro-Scan (.eeg) files were exported to standardized European Data Format (.edf) and subsequently transformed into structured tabular matrices (.csv) comprising 168,567 total time-series sample points across 19 active electrode channels and 1 binary target label.

TABLE I: CLINICAL DATASET SPECIFICATIONS & ACQUISITION PARAMETERS

Parameter / Attribute	Specification / Value	Data Type & Representation
Clinical Acquisition Center	Medicine Dept., Khulna Medical College & Hospital	Primary Hospital Cohort (Bangladesh)
Initial Screened / Final Balanced Cohort	51 Screened Patients -> 32 Curated Subjects	16 Seizure-Positive (1), 16 Negative (0)
Electrode Montage (10–20 System)	19 Scalp Channels (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T7, T8, P7, P8, Fz, Pz, Cz)	Continuous Multivariate Time-Series (Float)
Session Recording Duration	~1.5 Hours per Subject (Eyes Open / Closed)	Sequential Physiological Waveform
Total Time-Series Data Points	168,567 Sample Observations	Numerical Matrix (.eeg -> .edf -> .csv)
Missing / Null Values Post-Filtering	0 (Verified via Pandas DataFrame Audit)	Clean / Artifact-Attenuated

B. Multi-Band Bandpass Filtering in MATLAB (EEGLAB)

Raw scalp EEG signals are inherently contaminated by low-frequency baseline drift, ocular blink artifacts, and high-frequency power-line or electromyographic (EMG) interference. To isolate neurophysiologically meaningful oscillations and eliminate out-of-band noise, all .edf recordings were imported into MATLAB using the EEGLAB (v2021.1) signal processing toolbox. High-pass and low-pass attenuation were combined with bandpass filtering to decompose each 19-channel recording into three distinct neural frequency bands [13], [22]:

- **Alpha Band (α, 8–13 Hz):** Captures posterior dominant resting rhythms and thalamocortical synchronization, which frequently exhibit abnormal slowing or desynchronization in epileptic cohorts.
- **Beta Band (β, 14–29 Hz):** Reflects active sensorimotor and cortical excitability patterns associated with ictal and interictal spike-wave discharges.
- **Gamma Band (γ, 30–50 Hz):** Isolates high-frequency cortical oscillations strongly linked to localized epileptogenic zone recruitment and rapid neuronal firing.

Following spectral decomposition in EEGLAB, each filtered sub-band dataset was exported to .csv format and audited in Python (Pandas v3.9.11) to verify zero missing values, zero duplicate channels, and zero constant or quasi-constant attributes across all 19 electrode channels.

IV. FEATURE EXTRACTION & CLASSIFICATION METHODOLOGY

A. Statistical Time-Domain Biomarkers

To reduce dimensionality while preserving diagnostic separability across the 19 EEG channels, six mathematical time-domain features were extracted independently from the Alpha, Beta, and Gamma filtered signals. Let $x = \{x_1, x_2, ..., x_N\}$ denote a discrete EEG channel time-series of length N with sample mean $\bar{x}$ and sample standard deviation s_x:

1) Coefficient of Variation (CV): Quantifies the relative dispersion of EEG signal amplitude across time, aiding in the detection of transient burst variability [1]:

$$\bar{x} = (1 / N) \Sigma x_i \, , \quad s_x = sqrt([1 / (N - 1)] \Sigma (x_i - \bar{x})^2) \, , \quad CV = (s_x / \bar{x}) \times 100 \qquad \textbf{(1)}$$

2) Kurtosis (K): Measures the fourth-order standardized central moment of the amplitude distribution. High kurtosis captures sharp, high-amplitude epileptiform spikes, whereas low kurtosis reflects flatter background activity [41]:

$$K = [\Sigma (x_i - \bar{x})^4] / (N \cdot s_x^4) \qquad \textbf{(2)}$$

3) Skewness (S): Quantifies the third-order asymmetry of the EEG potential distribution around its mean, capturing polarity imbalances during spike-and-wave discharges [40]:

$$S = [\Sigma (x_i - \bar{x})^3] / [(N - 1) \cdot s_x^3] \qquad \textbf{(3)}$$

4) Hjorth Parameters (Activity, Mobility, Complexity): Introduced by Bo Hjorth for quantitative time-domain EEG characterization [10], [43], these three coupled indicators evaluate signal power surface, mean frequency, and bandwidth change:

$$Activity = var(x(t)) \, , \quad Mobility = sqrt(var(dx(t)/dt) / var(x(t))) \, , \quad Complexity = Mobility(dx(t)/dt) / Mobility(x(t)) \qquad \textbf{(4)}$$

5) Entropy & Signal Energy (Axial): Shannon/spectral entropy quantifies the irregularity, complexity, and predictability of nonlinear cortical oscillations, while Axial Energy ($E = \Sigma |x_i|^2$) captures cumulative paroxysmal discharge power across each channel [25].

B. Time-Frequency Spectrogram Representation & 2D CNN

To complement 1D statistical biomarkers with a time-frequency view, raw EEG recordings were transformed into 2D color spectral density arrays (spectrograms) plotting time on the horizontal axis, frequency on the vertical axis, and spectral power density via RGB pixel intensity [3], [38]. These 3-channel RGB spectrogram images were fed into a 2D Convolutional Neural Network (CNN) utilizing 3×3 convolutional kernels, nonlinear activation maps, and max-pooling layers to automatically extract spatial-spectral representations for binary classification [28], [42].

C. Supervised Classifiers, Dataset Partitioning & Hyperparameter Tuning

The 32-subject clinical dataset was partitioned using a stratified 80:20 train-test split, allocating 25 subjects for model training and hyperparameter optimization and 7 unseen subjects for holdout evaluation. Two interpretable tree-based classifiers were implemented in Python (scikit-learn):

• **Decision Tree (CART):** Constructs a hierarchical binary tree by recursively partitioning the feature space using impurity criteria—specifically Gini Impurity and Information Gain (Entropy)—until terminal leaf nodes predominantly represent a single diagnostic class [36].

• **Random Forest (RF) Ensemble:** An ensemble bagging architecture that constructs $B \in \{100, 200, 300\}$ decorrelated decision trees trained on bootstrap samples and random feature subspaces, aggregating predictions via majority voting to reduce variance and prevent overfitting on high-dimensional 19-channel inputs [37].

TABLE II: GRIDSEARCH HYPERPARAMETER SPACE & OPTIMAL TUNED CONFIGURATIONS (ALPHA, BETA & GAMMA BANDS)

Classifier	Hyperparameter	Search Space	Optimal Alpha (α) Config*	Optimal Beta (β) Config*	Optimal Gamma (γ) Config*
Decision Tree (CART)	max_features min_samples_leaf min_samples_split criterion max_depth	['auto', 'sqrt', 'log2'] [1, 2, 4] [2, 5, 10] ['gini', 'entropy'] [None, 10, 20]	log2 (Skew) / sqrt (Ent) 4 (Skew) / 1 (Ent) 5 (Skew) / 10 (Ent) gini (Skew, Ent) None (Skew, Ent)	auto (CV) / sqrt (Energy) 1 (CV) / 4 (Energy) 5 (CV) / 5 (Energy) gini (CV, Energy) 10 (CV, Energy)	auto (Hjorth) / sqrt (CV) 4 (Hjorth) / 1 (CV) 10 (Hjorth) / 2 (CV) gini (Hjorth) / entropy (CV) 20 (Hjorth) / 10 (CV)
Random Forest (Ensemble)	max_features min_samples_leaf min_samples_split n_estimators max_depth	['auto', 'sqrt', 'log2'] [1, 2, 4] [2, 5, 10] [100, 200, 300] [None, 10, 20]	sqrt (Skew) / auto (Hjorth, Ent) 2 (Skew, Hjorth, Ent) 5 (Skew, Hjorth) / 2 (Ent) 300 (Skew, Ent) / 100 (Hjorth) None (Skew, Ent) / 20 (Hjorth)	sqrt (Skew) / auto (Hjorth) 1 (Skew) / 2 (Hjorth) 2 (Skew, Hjorth) 300 (Skew) / 100 (Hjorth) 20 (Skew, Hjorth)	auto (Energy, Ent, Skew) 1 (Energy, Ent, Skew) 2 (Energy, Ent, Skew) 100 (Energy, Ent) / 300 (Skew) 20 (Energy, Ent) / 10 (Skew)

**Note: Optimal settings shown correspond to the top-performing biomarkers in each respective frequency band (Full 6-feature parameter grids are maintained in the repository).*

V. EXPERIMENTAL RESULTS & DISCUSSION

A. Evaluation Metrics

To rigorously evaluate diagnostic performance on the clinical epilepsy dataset, models were assessed using Confusion Matrix components—True Positives (TP: seizure-positive patients correctly diagnosed with epilepsy), True Negatives (TN: non-epileptic controls correctly identified as healthy), False Positives (FP: Type I error where healthy controls are misclassified as epileptic), and False Negatives (FN: Type II error where epileptic patients are missed, representing the most critical clinical risk in neurological screening) [5], [37]. From these quantities, Classification Accuracy, Precision (Positive Predictive Value), Recall (Sensitivity / True Positive Rate), F1-Score, and Area Under the Receiver Operating Characteristic Curve (ROC-AUC) were computed:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN), \quad Precision = TP / (TP + FP), \quad Recall = TP / (TP + FN), \quad F1 = (2 \cdot Precision \cdot Recall) / (Precision + Recall) \quad \textbf{(5)}$$

B. Exploratory Statistical & Inter-Channel Correlation Analysis

Prior to classification, box-plot distributions, histograms, and 19×19 Pearson correlation heatmaps were generated for all six biomarkers across the Alpha, Beta, and Gamma bands. Across all three frequency bands, Entropy Axial values exhibited tightly bounded Gaussian-like distributions centered between 12.0 and 12.5 bits with high inter-channel coherence ($r = 0.70–0.98$ across frontocentral and parieto-occipital pairs), reflecting global cortical regularity. Conversely, Kurtosis and Coefficient of Variation displayed low inter-channel correlation ($r < 0.30$ across most non-adjacent electrodes) and pronounced positive outliers on focal channels (notably C4-A2, O1-A1, T8-A2, and Fz-A1), confirming that higher-order moments capture localized, channel-specific paroxysmal spike bursts.

C. Multi-Band Feature-by-Feature Classification Performance

Tables III and IV summarize the baseline and hyperparameter-tuned classification performance across all six statistical features for the Alpha, Beta, and Gamma frequency bands. Several critical experimental findings emerge:

TABLE III: FEATURE-BY-FEATURE CLASSIFICATION PERFORMANCE ACROSS ALPHA, BETA, AND GAMMA FREQUENCY BANDS

Extracted Biomarker	Alpha (8–13 Hz) RF Base / Tuned	Alpha (8–13 Hz) DT Base / Tuned	Beta (14–29 Hz) RF Base / Tuned	Beta (14–29 Hz) DT Base / Tuned	Gamma (30–50 Hz) RF Base / Tuned	Gamma (30–50 Hz) DT Base / Tuned
Coefficient of Variation (CV)	71.0% / 83.0%	71.0% / 84.0%	71.0% / 83.0%	86.0% / 88.0%	71.0% / 89.0%	71.0% / 84.0%
Entropy Axial	71.0% / 86.0%	71.0% / 81.0%	71.0% / 86.0%	43.0% / 81.0%	86.0% / 89.0%	71.0% / 81.0%
Skewness	86.0% / 89.0%	71.0% / 85.0%	86.0% / 89.0%	86.0% / 83.0%	86.0% / 89.0%	57.0% / 83.0%
Kurtosis	57.0% / 84.0%	71.0% / 83.0%	57.0% / 84.0%	71.0% / 81.5%	71.0% / 88.0%	43.0% / 84.0%
Hjorth Parameters	71.0% / 88.0%	71.0% / 84.0%	71.0% / 88.0%	57.0% / 84.0%	43.0% / 89.0%	71.0% / 89.0%
Energy Axial	43.0% / 81.0%	43.0% / 81.0%	43.0% / 81.0%	71.0% / 85.5%	71.0% / 90.0%	71.0% / 83.0%
Peak Band Performance	**90.0% (AUC 0.95)**	**84.0% (AUC 0.58)**	**89.0% (AUC 0.93)**	**88.0% (AUC 0.60)**	**90.0% (AUC 0.83)**	**85.0% (AUC 0.58)**

First, GridSearch hyperparameter tuning yielded substantial performance gains across every feature and frequency band, improving Random Forest accuracy from 43.0%–86.0% (untuned) to 81.0%–90.0% (tuned). Second, within the Alpha band (8–13 Hz), Random Forest achieved an overall peak accuracy of 90.0% and the highest discriminative separability in the study with an ROC-AUC of 0.95 (driven by Skewness at 89%, Hjorth Parameters at 88%, and Entropy Axial at 86% with 100% sensitivity on seizure-positive cases). Third, in the Gamma band (30–50 Hz), where high-frequency epileptogenic discharges concentrate power, Energy Axial with Random Forest reached 90.0% tuned accuracy, while Entropy, Skewness, Hjorth Parameters, and CV all achieved 89.0% tuned accuracy. Fourth, on the Beta band (14–29 Hz), Random Forest achieved 89.0% accuracy and a strong ROC-AUC of 0.93, closely followed by Decision Tree at 88.0% on Coefficient of Variation.

By contrast, the 2D CNN trained directly on time-frequency spectrogram images achieved a validation accuracy of 75.0%. Inspection of the epoch-wise training and validation accuracy curves (10 epochs) confirms that while spectrogram CNNs capture visual frequency banding, deep convolutional architectures require significantly larger sample sizes than N = 32 subjects to generalize without high validation variance, whereas Random Forest ensembles excel on structured multi-band statistical biomarkers.

TABLE IV: CLASS-WISE DIAGNOSTIC METRICS FOR TOP MODELS & BENCHMARK COMPARISON WITH PRIOR LITERATURE

Study / Model Configuration	Frequency Band / Modality	Classifier Architecture	Precision (Pos / Neg)	Recall (Pos / Neg)	F1-Score (Pos / Neg)	Overall Accuracy / AUC
Fraiwan et al. (2020) [3]	Broadband EEG Signals	Random Forest	—	—	—	75.0%
Wang et al. (2021) [6]	Filtered EEG Features	Support Vector Machine (SVM)	—	—	—	82.0%
Chiang et al. (2019) [4], [10]	Wavelet / Fuzzy Entropy	Decision Tree / Associative Net	—	—	—	85.39%
Proposed Spectrogram Baseline	2D Time-Frequency Spectrogram	2D Convolutional Neural Net (CNN)	—	—	—	75.0%
Proposed Alpha-Band Model	Alpha Band (8–13 Hz, Skewness/Ent)	Tuned Random Forest (300 Trees)	75% / 100%	100% / 75%	86% / 86%	**90.0% (AUC = 0.95)**

Proposed Beta-Band Model	Beta Band (14–29 Hz, Skewness/Hjorth)	Tuned Random Forest (300 Trees)	100% / 60%	67% / 100%	75% / 80%	**89.0% (AUC = 0.93)**
Proposed Gamma-Band Model	Gamma Band (30–50 Hz, Energy/Ent)	Tuned Random Forest (100 Trees)	100% / 100%*	67% / 100%*	80% / 89%*	**90.0% (AUC = 0.83)**

VI. CONCLUSION, LIMITATIONS & FUTURE WORK

This study developed and validated an end-to-end clinical machine learning pipeline for automated epilepsy detection using 19-channel primary EEG recordings acquired from 32 patients at Khulna Medical College and Hospital. By decomposing raw clinical EEG signals into Alpha (8–13 Hz), Beta (14–29 Hz), and Gamma (30–50 Hz) frequency bands in MATLAB (EEGLAB) and independently evaluating six time-domain biomarkers alongside 2D spectrograms, we demonstrated that hyperparameter-tuned Random Forest classifiers achieve 90.0% accuracy (0.95 ROC-AUC) in the Alpha band, 89.0% accuracy (0.93 ROC-AUC) in the Beta band, and 90.0% accuracy (0.83 ROC-AUC) in the Gamma band. Separating feature evaluation by frequency band confirmed that Skewness, Hjorth Parameters, Entropy Axial, and Gamma-band Energy Axial provide the strongest diagnostic separability for clinical seizure screening.

The primary limitation of the present study is the sample size of the balanced clinical cohort (N = 32 subjects), which constrained the generalization capacity of the 2D CNN spectrogram model (75.0% accuracy) and introduced step-wise discretization in holdout confusion matrices. In future work, we aim to: (1) expand multi-center clinical EEG data collection across hospitals in Bangladesh, (2) evaluate subject-independent Leave-One-Subject-Out (LOSO) cross-validation with multi-band feature fusion, and (3) investigate lightweight hybrid CNN-Transformer and Graph Neural Network (GNN) architectures that model spatial electrode topology across the 19-channel 10–20 montage.

VII. ACKNOWLEDGMENT

The authors express their sincere gratitude to Assistant Professor Zahrul Jannat Peya and the faculty of the Department of Computer Science and Engineering at North Western University, Khulna, for their supervision and technical guidance, as well as the clinical staff of the Medicine Department at Khulna Medical College and Hospital for supporting primary EEG data acquisition.

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